March 15, 2025

VIA EMAIL to [redacted]

AI Action Plan
Attn: Faisal D'Souza, NCO
2415 Eisenhower Ave.
Alexandria, VA 22314

Re: AI Action Plan

In its "Request for Information on the Development of an Artificial Intelligence (AI) Action Plan," 90 Fed. Reg. 9088 (Feb. 6, 2025), the White House Office of Science and Technology Policy (OSTP) requested input from interested parties on the development of an AI Action Plan that will define the priority policy actions needed to maintain America's AI dominance. I am a professor at Florida International University and a Senior Research Fellow in the Center for Energy, Climate, and Environment at the Heritage Foundation. From 2017-2019 I was associate director for regulatory reform at the White House Council on Environmental Quality. I respectfully submit these comments for your consideration.*

One focal point of my work is what America needs to do to maintain its global dominance in high technology. So far in the 21st century, America has managed to maintain its edge in the most advanced technologies. But China is catching up fast, and its progress is accelerating at an alarming rate in every domain. In the areas of greatest U.S. competitive advantage, such as the most sophisticated design and engineering, China is quickly closing the gap, while in the areas of greatest Chinese competitive advantage, such as economies of scale and low production cost, the Chinese lead is only widening. Meanwhile, China is overcoming the obstacles to its technological progress faster than America is overcoming the obstacles that it faces.

The 21st century can be another century of American technological dominance—but only if we can escape the stifling regulation that drive investment and jobs offshore and embrace once again the fearless competition that made American industry great the first time. As Vice President J.D. Vance said at the Paris AI Action summit on February 11, 2025, "AI will have countless revolutionary applications. . . . And to restrict its development now would not only unfairly benefit incumbents in this space, it would mean paralyzing one of the most promising technologies we have seen in generations."

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That is nowhere more true than in the short sprint to Artificial General Intelligence.

Introduction: The Sprint to AGI

President Trump has defined America's objective in the AI race: "[T]o sustain and enhance America's global AI dominance in order to promote human flourishing, economic competitiveness, and national security." Executive Order 14179, "Removing Barriers to American Leadership in Artificial Intelligence," (Jan. 23, 2025).

The race for multi-generational AI dominance may be determined in just the next few years by which country is first to develop a commercially successful ecosystem for Artificial General Intelligence (AGI).[*] AGI is likely to be one of the most revolutionary technologies in the history of civilization. As a disruptive innovation, it will entail new risks and challenges for some groups, which we must be mindful of. But AGI also has incalculable potential for enhancing wealth creation, labor productivity, quality of life, and national power.

Given the revolutionary potential of AGI, the competition for the 21st century could be determined in large part by which countries get to AGI first. In the military domain, that country will be able to achieve both a strategic offensive advantage and the ability to defend against adversaries bent on developing AGI themselves. As America did with the internet, that country will have the opportunity to define global best practices in both commerce and regulation, and shape global ethical standards, thereby influencing international norms. In short, the first nation to achieve AGI will realize enormous gains in military, technological, economic, and social power, thereby maximizing the benefits and helping to mitigate the risks of AI.

The AI Action Plan should take as point of departure a systematic assessment of the comparative advantages and disadvantages that the U.S. and China each have in the race to AGI. It should seek to build on America's comparative advantages by removing or reducing obstacles to innovation here at home, while improving the use of national security authorities to slow adversaries' development of technologies with military applications.

Policies must be judged on their likely results, not their intentions. Every regulation that affects America's high-tech sector must be judged on the basis of its comparative costs and benefits.

Investment depends upon the competitiveness of a country's framework of regulation and taxation. The AI Action Plan should prioritize the lowering of obstacles to investment and

[*] AGI can be defined as AI capable of performing any intellectual task that a human being can do, with human-level understanding, flexibility, adaptability, and generality.

innovation, with the goal of keeping America's AI ecosystem the most commercially competitive in the world. Staying on the cutting edge of the AI revolution will require a massive amount of compute capacity. Indeed, the amount of compute that AI requires—and the amount of energy needed—is rising exponentially with ever step forward in AI. With that in mind, the AI Action Plan should focus on the following priorities:

- Ensure that key production factors—such as skilled labor, key components, raw materials, and especially electricity—are available in abundance and at globally competitive prices.

- Ensure that national security and other policy priorities are addressed effectively and without unfairly benefiting incumbents or resulting in other unintended consequences that might undermine the purpose of the policy.

- Ensure a proper distribution of federal and state regulatory authorities that allows AI firms to develop economies of scale within a seamless national market while states regulate the use of AI within their borders in accordance with local preferences.

With those recommended policy priorities, this letter focuses on four key action items relating to (1) energy, (2) AI diffusion control, (3) division of regulatory authorities among federal and state governments, and (4) the overriding need for a competitive framework of regulation so American workers and companies can compete globally.

1. America's AI sector urgently needs significant additional fossil and nuclear electricity capacity.

America's AI revolution is facing an immediate danger that policymakers are only starting to appreciate. If the barriers to rapid expansion in new baseload electrical generation capacity (coal, natural gas, and nuclear) and transmission are not removed very quickly, the AI revolution could quickly move offshore in search of affordable power. That is because the power requirements of AI are soaring, but the U.S. is facing severe constraints on the ability of its electricity grid to meet the new demand.

The power requirements of AI training and inference and growing exponentially with every new advance in AI. According to Semianalysis, "Global Datacenter Critical IT power demand will surge from 49 Gigawatts (GW) in 2023 to 96 GW by 2026, of which AI will consume ~40 GW."[*] 40 GW is the equivalent of 40 average-size nuclear plants. For reference there are currently under 100 nuclear plants in operation in the U.S., less than we had in 1980.

[*] https://semianalysis.com/2024/03/13/ai-datacenter-energy-dilemma-race/

Under current law, it seems increasingly unlikely that most of that new power demand can be met in America. The U.S. power sector is approaching an unprecedented capacity shortfall. A major reliability crisis is now clearly looming. The latest assessment from the North American Electric Reliability Corporation (NERC) is particularly alarming.[*] The combination of soaring demand and rapid retirements of coal plants—approximately 15% of the current U.S. generation capacity is expected to retire by 2030—with slow and insufficient deployment of replacement generation, adds up to a capacity shortfall of at least 30% of power generation capacity by 2030.

Part of the problem is that real baseload generation is being replaced by a renewable energy mirage. In 2024 alone, 8.45 GW of coal capacity (equivalent to nine nuclear plants) was retired, yet new natural gas capacity barely offset nuclear retirements. Although significant solar (13 GW) and battery (4 GW) capacity was added, their effective contributions are much less than nominal capacity due to low capacity factors (around 24% for solar, and even less during critical winter demand peaks), and solar can't be relied upon to meet even those lower capacity ratings when needed. Consequently, NERC anticipates serious electricity shortfalls starting this summer in Midwestern states, highlighting that resource additions are far below what's required to meet rising demand.

As the NERC report shows, while electrical generation capacity appears to be expanding in the U.S. in terms of nominal capacity, it is in fact contracting in terms of projected power availability, in the face of soaring demand. More than half of the world's hyperscale data center capacity—and the vast majority of its AI compute—are currently located in the U.S. But with electricity prices already 30% higher than in 2020 and forecast to rise further in the face of the looming capacity shortfall, data center operators are increasingly likely to look abroad for affordable and reliable electricity. This is why President Trump's emphasis on energy dominance is so important.

A combination of anti-energy policies has created severe constraints on the ability of generation capacity to keep pace with growing demand, chiefly by creating prohibitive risks and obstacles for private investment in the power plants we actually need. The administration should move as quickly as possible to address these issues:

- *Inflation Reduction Act subsidies for renewable energy projects*. IRA subsidies create an unfavorable revenue structure for baseload generators (coal, natural gas, and nuclear) which can't fully recoup costs with all the subsidized renewable energy getting dumped

[*] NERC Long-Term Reliability Assessment (Dec. 2024), https://www.nerc.com/pa/RAPA/ra/Reliability%20Assessments%20DL/NERC_Long%20Term%20Reliability%20Assessment_2024.pdf

on the grid. That is killing investment in the generation capacity we actually need. Congress should repeal the IRA as soon as possible, as every day that passes with the subsidies in place creates a deeper hole for America's electricity grid.

- *Unfavorable electricity market structure*. At the state level, a combination of policies have created further constraints on the ability of electrical capacity to meet new demand. State renewable portfolio standards (RPS) and "net zero" carbon emission mandates penalize baseload generation. Meanwhile, the lack of adequate reliability standards among Regional Transmission Organizations (RTOs) has made the interaction of IRA subsidies and RTO market structure particularly toxic for investment in baseload capacity. The administration should encourage states to unwind the RPS and "net zero" mandates, and should work with FERC and Congress to modernize the RTO market structure so that it prioritizes electrical reliability.

- *Inefficient Permitting.* The costs, risks, and uncertainties of the federal process for permitting and environmental review of major infrastructure projects are perhaps the worst in the developed world. Projects that take just months to permit in other developed countries, take years, and cost several times more, to permit in the U.S. In accordance with the Unleashing American Energy Executive Order,[*] the administration should propose sweeping legislative reforms that create a one-stop-shop permitting agency and single permit application process for critically needed energy infrastructure, with provisional permit decisions guaranteed in a year or less.

2. **Any export control regime on AI Diffusion must be carefully assessed for efficacy and unintended consequences.**

On Jan. 15, 2025, the Department of Commerce, Bureau of Industry and Security (BIS) released a "Framework for Artificial Intelligence Diffusion." 90 Fed. Reg. 4544. The Framework is extremely ambitious and will have enormous effect on where data centers are built around the world in the years to come and whether they use American or Chinese technology.

My colleagues and I are studying the Framework carefully and expect to file detailed comments to BIS by the May 15, 2025, deadline. At this juncture, I would offer the following preliminary questions:

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[*] Executive Order 14154, "Unleashing American Energy," (January 20, 2025), instructs the heads of the National Economic Council and Office of Management and Budget to make legislative proposals on permitting reform to Congress.

- *Does the Framework effectively address the national security risks of AI?* According to BIS, "AI models have the potential to enable advanced military and intelligence applications; lower the barriers to entry for non-experts to develop weapons of mass destruction (WMD); support powerful offensive cyber operations; and assist in human rights violations, such as through mass surveillance." In a strong rule, BIS would explain how the Framework will prevent or delay these risks from materializing. With respect to advanced military/intelligence capabilities, powerful cyber operations, and mass surveillance, for example, what exactly might the Chinese Communist Party be able to do with access to the most advanced GPUs and models that it can't do now? This became a pressing question with the release of DeepSeek R1, a high-performance, ultra-low-sparsity frontier AI model, by a company based in Zheijiang, China, just weeks after the Framework's publication.

- *Is the Framework more like an export control on the avionics of a stealth bomber, or is it more like an export control on the transmission of a Ford Motel T?* If the former, an export control would be desirable. If the latter, then the risk of minimal negative impact on Chinese developers combined with significant negative impact on the export competitiveness of American manufacturers should be considered.

3. A pro-innovation regulatory framework should balance interstate regulatory competition with the need for a seamless national market.

In general, the U.S. should take the same approach to regulating AI that it took to regulating the internet from its inception. Like the internet, cloud computing, including AI models, are an instrumentality of interstate commerce, and are properly the domain of federal regulators. State legislation that undermines federal regulatory authority in this area should be explicitly preempted if not already barred on "dormant Commerce Clause" grounds. On the other hand, states should be free to regulate the use of AI in accordance with local preferences.

As with the internet, this distribution of regulatory authorities acknowledges that American leadership in AI requires the development of a seamless national market under uniform federal standards in order to foster innovation and maintain competitiveness. Establishing a uniform national framework that reduces regulatory barriers is crucial for creating a cohesive national market and achieving economies of scale. Federal preemption in AI model development prevents a fragmented regulatory environment that could hinder innovation.

The right allocation of regulatory authorities also acknowledges the importance of local choice. Federal regulations should establish baseline standards for data protection and ethical AI development, ensuring that AI systems are designed with privacy considerations and that users can meaningfully choose their desired level of privacy. The right framework will also leave individual states to regulate the use of AI within their jurisdictions. That allows states and local communities to address ethical and safety issues, such as child safety, effectively.

States are best positioned to address specific concerns regarding AI use. They can enact laws that reflect regional values and priorities, such as prohibiting certain AI applications deemed harmful or unethical within their territories. Allowing states to regulate the use of AI ensures that local risks are mitigated in accordance with existing state laws, fostering public trust and acceptance of AI technologies.

Congress and the administration should distinguish carefully between different categories of AI regulation at the state level, some of which are desirable and some of which have the potential to be deeply problematic.

- *Preexisting regulations with incidental application to AI.* Many of the perceived risks of AI are already effectively mitigated under general legal principles, such as tort and criminal law. For instance, if AI generates defamatory content, and that content is propagated by users, the responsible users are subject to defamation laws. Similarly, using AI to harm children can make the user criminally liable under existing law.

- *Regulations on how AI is used.* Some state bills regulate the use of AI to create deepfakes and otherwise require transparency with respect to the origin of content generated by AI. State regulatory innovation with respect to novel problems that arise in the use of AI is important and should be given broad latitude.

- *Regulations on how foundation models are trained and structured.* This is the riskiest category of state regulation, and unfortunately also the one that appears to have seen the largest number of state bills filed. For example, a recent AI safety bill in California, SB 1074, would have made AI companies liable for harms caused by their foundation models. Gov. Gavin Newsom vetoed the bill in September 2024, but only after it passed with broad support in the California legislature, and other states have moved in a similar direction. There is potentially an argument that such bills violate the "dormant Commerce Clause" by interfering with the federal government's power to regulate interstate commerce, and Congress should consider expressly preempting such bills.

4. The imperative of a competitive climate of regulation and taxation for U.S. industry.

America's private sector remains the world's most innovative and productive. But a century of progressive government, in the form of stifling regulation, has driven companies and jobs offshore. As the blight of Rust Belt towns, Appalachian communities, and inner cities attests, the consequences of overregulation can persist for decades.

Investment flows where taxes and regulations are low and production factors like labor and electricity are reliable and affordable. In all those metrics the U.S. is falling further and further behind much of the world. Even in America's most innovative sectors—like high technology—warning signs are everywhere. The entire supply chain for semiconductor manufacturing has moved offshore, with only high-end engineering remaining in the U.S. China is already making inroads into these areas, and if America doesn't act fast, it could soon start falling behind even in the high-tech race.

The AI Action Plan has the greatest chance of succeeding if it is part of a national commitment to make America once again the most attractive country in the world in which to invest and work. That means sweeping regulatory reforms and embracing once again the permissionless innovation that made America the world's technological leader in the last hundred years.

Conclusion

The United States is today engaged in a race against China for dominance in high technology. This competition will determine whether, in the century ahead, the world remains safe for the freedom and prosperity on which Americans vitally depend. The central theater of that competition will almost certainly be AI and particularly the development of AGI.

As Vice President Vance said, America now has the chance to "catch lightning in a bottle." But there is no time to waste.

Thank you for your consideration of these comments.

Respectfully submitted,

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* These comments represent my views and not necessarily those of Florida International University or the Heritage Foundation